Linear Models Lecture 12: Hypothesis Testing

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March 6, 2026

Where We Are

Last lecture: We derived the Wald statistic

$$W = (\boldsymbol{R}\hat{\boldsymbol{\beta}} - \boldsymbol{r})' \left[\boldsymbol{R}\hat{\boldsymbol{V}}\boldsymbol{R}'\right]^{-1} (\boldsymbol{R}\hat{\boldsymbol{\beta}} - \boldsymbol{r}) \xrightarrow{d} \chi^2_q$$

and showed how to construct robust tests and confidence intervals.

Today: We go deeper into the *practice* of hypothesis testing.

- The F test: a criterion-based alternative
- Score (LM) tests: testing from the restricted model
- The trinity of classical tests
- Test inversion for confidence regions
- Multiple testing and Bonferroni corrections
- Power: what determines whether you can detect an effect?
- Hansen's practical advice for applied work

Constrained Least Squares: Setup

Many tests today compare an **unrestricted** model to a **restricted** one. We need to know how to estimate under restrictions.

Problem: Minimize the sum of squared errors subject to q linear constraints:

$$\tilde{\beta}_{\text{CLS}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} (Y_i - \boldsymbol{X}_i'\boldsymbol{\beta})^2 \quad \text{subject to} \quad \boldsymbol{R}'\boldsymbol{\beta} = \boldsymbol{r}$$

where $\boldsymbol{R}$ is $k \times q$ and $\boldsymbol{r}$ is $q \times 1$.

Examples:

- $\beta_3 = 0$: set $\boldsymbol{R}' = (0, 0, 1, 0, \ldots)$, $\boldsymbol{r} = 0$ (exclusion restriction)
- $\beta_2 = \beta_3$: set $\boldsymbol{R}' = (0, 1, -1, 0, \ldots)$, $\boldsymbol{r} = 0$ (equality restriction)
- $\beta_2 + \beta_3 = 1$: set $\boldsymbol{R}' = (0, 1, 1, 0, \ldots)$, $\boldsymbol{r} = 1$ (adding-up constraint)

The CLS Estimator

Solve via Lagrange multipliers. The solution is:

$$\boxed{\tilde{\beta}_{\text{CLS}} = \hat{\beta}_{\text{OLS}} - (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}\left[\boldsymbol{R}'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}\right]^{-1}(\boldsymbol{R}'\hat{\beta}_{\text{OLS}} - \boldsymbol{r})}$$

Reading the formula:

- Start from unrestricted OLS $\hat{\beta}_{\text{OLS}}$, subtract a correction toward the constraint
- Correction is proportional to $(\boldsymbol{R}'\hat{\beta}_{\text{OLS}} - \boldsymbol{r})$: how far OLS violates H_0
- If OLS already satisfies the restriction, $\tilde{\beta}_{\text{CLS}} = \hat{\beta}_{\text{OLS}}$

The restricted residuals and variance estimate are:

$$\tilde{e}_i = Y_i - \boldsymbol{X}_i'\tilde{\beta}_{\text{CLS}}, \qquad \tilde{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\tilde{e}_i^2$$

Since the restriction constrains the parameter space: $\tilde{\sigma}^2 \geq \hat{\sigma}^2$ always.

CLS in Practice

Simple case: If H_0 sets some coefficients to zero, CLS just means dropping those regressors.

```
# Unrestricted
mod_U <- lm(lwage ~ education + exper + I(exper^2) +
              female + union, data = wages)
# Restricted (H0: beta_female = beta_union = 0)
mod_R <- lm(lwage ~ education + exper + I(exper^2),
              data = wages)
```

General case: For arbitrary linear restrictions (e.g., $\beta_2 = \beta_3$), reparameterize or use the closed-form formula.

The key outputs from CLS are $\text{SSE}_R = \sum \tilde{e}_i^2$ and $\tilde{\sigma}^2$, which we use to build the F and Score statistics.

The F Test: Idea

The F test asks: **how much does the fit worsen when we impose the null?**

- Run the **unrestricted** regression: get $\text{SSE}_U = \sum(Y_i - \boldsymbol{X}_i'\hat{\boldsymbol{\beta}}_{\text{OLS}})^2$
- Run the **restricted** regression (imposing H_0): get $\text{SSE}_R = \sum(Y_i - \boldsymbol{X}_i'\tilde{\boldsymbol{\beta}}_{\text{CLS}})^2$

- Since H_0 constrains the parameter space, $\text{SSE}_R \geq \text{SSE}_U$ always.
- If the fit barely worsens $\Rightarrow$ the restriction is consistent with the data.
- If the fit worsens a lot $\Rightarrow$ the restriction is costly $\Rightarrow$ evidence against H_0.

The F Statistic

Testing $H_0: \boldsymbol{R}'\boldsymbol{\beta} = \boldsymbol{r}$ with q restrictions:

$$\boxed{F = \frac{(\text{SSE}_R - \text{SSE}_U)/q}{\text{SSE}_U/(n-k)} = \frac{(\tilde{\sigma}^2 - \hat{\sigma}^2)/q}{\hat{\sigma}^2/(n-k)}}$$

- **Numerator**: average increase in residual variance per restriction
- **Denominator**: unrestricted estimate of error variance (s^2)
- Reject H_0 when F is large

Key relationship: For linear hypotheses, $F = W^0/q$ where W^0 is the homoskedastic Wald statistic. Under homoskedasticity and normality, $F \sim F_{q,\,n-k}$ exactly. Asymptotically, $F \xrightarrow{d} \chi^2_q/q$.

F Test: When and Why

Advantages:

- Directly computable from standard output (just need SSE from two regressions)
- Exact distribution under normality + homoskedasticity
- Slightly more conservative than χ^2 critical values (good in small samples)

Limitations:

- Requires homoskedasticity for the $F_{q,n-k}$ distribution to be valid
- Under heteroskedasticity, use the robust Wald test from Lecture 11 instead

Hansen's warning: Many packages automatically report an "F-statistic" testing that all slopes are zero. With modern sample sizes this is nearly always significant. **There is no reason to report this F statistic.**

F Test in R

```
mod_U <- lm(lwage ~ education + exper + I(exper^2) +
              female + union, data = wages)
mod_R <- lm(lwage ~ education + exper + I(exper^2),
              data = wages)

# Manual F test
SSE_U <- sum(resid(mod_U)^2)
SSE_R <- sum(resid(mod_R)^2)
q <- 2; n <- nobs(mod_U); k <- length(coef(mod_U))
F_stat <- ((SSE_R - SSE_U)/q) / (SSE_U/(n - k))
p_val  <- 1 - pf(F_stat, q, n - k)

# Or simply:
anova(mod_R, mod_U)
```

Example: Joint Significance

Wage regression with "Male Union Member" and "Female Union Member" indicators.

Test: H_0: Union membership has no effect on wages (both coefficients $= 0$).

- $W = 23$ (Wald), so $F = 23/2 = 11.5$, $p < 0.001$. **Reject.**

Interpretation: Rejecting the joint null means *at least one* coefficient is nonzero. It does not mean both are. Always examine both the joint test and individual t-statistics for a complete picture.

Score Tests: The Idea

The Wald test starts from the **unrestricted** estimate and asks: is $\hat{\boldsymbol{\theta}}$ far from $\boldsymbol{\theta}_0$?

The **Score test** (also called **Lagrange Multiplier test**) starts from the **restricted** estimate and asks: does the objective function want to move away from the restriction?

- Compute the restricted estimator $\tilde{\boldsymbol{\beta}}$ (constrained least squares)
- Evaluate the *gradient* (score) of the objective function at $\tilde{\boldsymbol{\beta}}$
- If H_0 is true, the score should be near zero (we're near the optimum)
- If H_0 is false, the score should be large (the restriction is pulling us away)

Score Statistic for Linear Restrictions

For H_0: $\boldsymbol{R}'\boldsymbol{\beta} = \boldsymbol{c}$ in the normal regression model:

$$S = \frac{(\boldsymbol{R}'\hat{\boldsymbol{\beta}} - \boldsymbol{c})'[\boldsymbol{R}'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{R}]^{-1}(\boldsymbol{R}'\hat{\boldsymbol{\beta}} - \boldsymbol{c})}{\tilde{\sigma}^2}$$

- Identical to W^0 except s^2 replaced by $\tilde{\sigma}^2$ (restricted variance).
- Under H_0: $S \xrightarrow{d} \chi^2_q$

Key result: S is a monotone transformation of F: $S = n\left(1 - \frac{1}{1+qF/(n-k)}\right)$, so the Score test and the F test always give the same accept/reject decision.

When Are Score Tests Useful?

For linear regression with linear restrictions, the Score test offers nothing new over F or Wald.

But in more complex settings, Score tests have a major advantage:

- They only require estimation under H_0 (the restricted model)
- The "unrestricted" model may not even be a simple OLS regression

Example: Testing for heteroskedasticity (Breusch–Pagan). The null is homoskedastic OLS—easy. The unrestricted model allows $\text{Var}(e_i \mid X_i)$ to vary with X, which requires modeling the variance function. The Score test only needs the OLS residuals.

Same idea applies to:

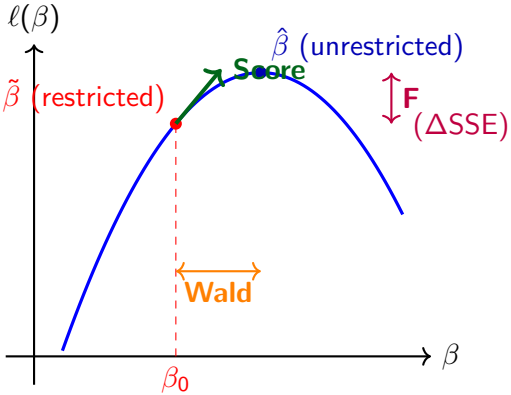
- Testing for serial correlation (Breusch–Godfrey): null is OLS with iid errors
- Testing for omitted nonlinearities: null is a simple linear specification
- Any setting where H_0 gives you a clean model but H_1 is messy

Three Ways to Test the Same Hypothesis

	Wald	**Score (LM)**	**F / LR-like**
Estimates from	Unrestricted	Restricted	Both
Measures	Distance of $\hat{\boldsymbol{\theta}}$ from $\boldsymbol{\theta}_0$	Gradient at restriction	Change in fit (SSE)
Null dist.	χ^2_q	χ^2_q	χ^2_q/q or $F_{q,n-k}$
Robust?	Yes (sandwich)	Score-like	No

For linear restrictions in the normal model, all three are monotone transformations of each other and give identical decisions. They differ for nonlinear restrictions or non-normal models.

Visualizing the Trinity



- **Wald**: horizontal distance between $\hat{\beta}$ and β_0 **Score**: slope at restricted estimate **F**: vertical distance (ΔSSE)

Confidence Intervals *Are* Inverted Tests

Recall the standard 95% CI:

$$\hat{C} = \left[\hat{\theta} - 1.96 \cdot s(\hat{\theta}),\ \ \hat{\theta} + 1.96 \cdot s(\hat{\theta})\right]$$

This is exactly the set of values θ that are *not rejected* by a two-sided t-test:

$$\hat{C} = \{\theta \ : \ |T(\theta)| \leq 1.96\}$$

General principle: Inverting a test with good Type I error control produces a confidence set with good coverage.

$$P[\theta \in \hat{C}] = P[\text{Accept} \mid \theta] = 1 - P[\text{Type I error}]$$

Why Test Inversion Matters: Nonlinear Parameters

Consider $\theta = \beta_1/\beta_2$ (e.g., peak experience in a wage equation).

Approach 1: Delta method CI

$$\hat{\theta} \pm 1.96 \cdot s(\hat{\theta})$$

From Lecture 11, this can be inaccurate for ratios (Fieller's problem).

Approach 2: Rewrite as a *linear* restriction $\beta_1 - \theta\beta_2 = 0$ and invert:

$$\hat{C} = \left\{ \theta \ : \ \frac{(\hat{\beta}_1 - \theta\hat{\beta}_2)^2}{\boldsymbol{R}'(\theta)\hat{\boldsymbol{V}}\boldsymbol{R}(\theta)} \leq 1.96^2 \right\}$$

where $\boldsymbol{R}(\theta) = (1, -\theta)'$. This requires a grid search over θ.

Hansen's example: Peak experience $= -50\beta_1/\beta_2$.
Delta method CI: [29.8, 29.9]. Inverted linear test CI: [29.1, 30.6].
The delta method interval is **far too narrow**.

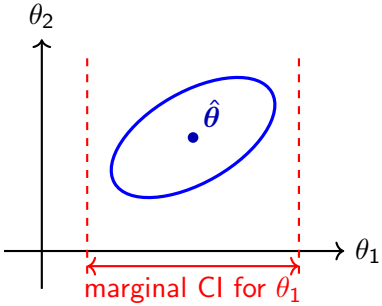
Test Inversion in R

```
# Wage equation: log(wage) ~ exper + exper^2/100 + ...
mod <- lm(lwage ~ exper + I(exper^2/100) + education,
          data = wages)
b <- coef(mod); V <- vcovHC(mod)

# Grid search: invert the t-test for theta = -50*b1/b2
theta_grid <- seq(20, 40, by = 0.01)
in_CI <- sapply(theta_grid, function(th) {
  R <- c(0, 1, -th/50, 0)  # gradient of b[2] - th*b[3]/50
  num <- (b[2] - th * b[3] / 50)^2
  den <- t(R) %*% V %*% R
  num / den <= qchisq(0.95, 1)
})
CI <- range(theta_grid[in_CI])
```

Confidence Regions for Multiple Parameters

For $q > 1$ parameters, invert the Wald test: $\hat{C} = \left\{ \boldsymbol{\theta} : W(\boldsymbol{\theta}) \leq \chi^2_{q,1-\alpha} \right\}$ — an **ellipsoid** in $\mathbb{R}^q$ centered at $\hat{\boldsymbol{\theta}}$, shaped by $\hat{\boldsymbol{V}}_\theta^{-1}$.



Marginal CIs (projections) are always wider than the ellipsoid in each dimension.

The Multiple Testing Problem

Suppose you test k hypotheses, each at level $\alpha = 0.05$. Under the **global null** (all k true), what is $P[\text{at least one false rejection}]$?

By **Boole's inequality**:

$$P\left[\min_{j \le k} p_j < \alpha\right] \le \sum_{j=1}^{k} P[p_j < \alpha] \to k\alpha$$

k (tests)	α per test	Familywise error $\le$
5	0.05	0.25
10	0.05	0.50
20	0.05	1.00

With 20 tests, you are *virtually certain* to get a false rejection!

The Bonferroni Correction

Goal: Control the *familywise error rate* (FWER) at α.

Rule: Reject the jth hypothesis only if $p_j < \alpha/k$. Equivalently: $p_{\text{Bonf}} = k \cdot \min_{j \leq k} \; p_j$.

Proof:

$$P\left[\min_{j \leq k} p_j < \frac{\alpha}{k}\right] \leq \sum_{j=1}^{k} P\left[p_j < \frac{\alpha}{k}\right] \rightarrow k \cdot \frac{\alpha}{k} = \alpha$$

Simple and conservative. Controls FWER for *any* dependence structure among the tests. The cost: reduced power when k is large.

Bonferroni: Worked Example

Two coefficients tested at $\alpha = 0.05$:

	Individual p	Bonferroni p ($= 2 \times p$)
Union membership	0.04	0.08
Married status	0.15	0.30

- **Without correction:** Union membership "significant" at 5%.
- **With Bonferroni:** Neither significant at FWER $= 0.05$ (need $p < 0.025$).

When to worry: exploring many specifications/subgroups, examining many coefficients, or reporting the "most significant" result.

Bonferroni in R

```
# Get p-values from a regression
mod <- lm(lwage ~ education + exper + I(exper^2) +
            female + union + married, data = wages)
pvals <- summary(mod)$coefficients[-1, 4]  # drop intercept

# Bonferroni correction
p_bonf <- p.adjust(pvals, method = "bonferroni")
cbind(raw = round(pvals, 4), bonferroni = round(p_bonf, 4))

# Other options: Holm (less conservative, still controls FWER)
p_holm <- p.adjust(pvals, method = "holm")
```

Holm's method is uniformly more powerful than Bonferroni while still controlling FWER. Use it when available.

Type I and Type II Errors

	H_0 **true**	H_0 **false**
Reject H_0	Type I error (α)	Correct (Power $= 1 - \beta$)
Accept H_0	Correct ($1 - \alpha$)	Type II error (β)

- **Size** (α): probability of falsely rejecting a true null
- **Power** (π): probability of correctly rejecting a false null: $\pi(\theta) = P[\text{Reject } H_0 \mid \theta \neq \theta_0]$
- Power depends on the *true value* of θ, on n, and on α.

Key trade-off: Making α smaller reduces Type I errors but also reduces power. A test with $\alpha = 0$ never rejects and has zero power.

Power of the t-Test

For $H_0 : \theta = \theta_0$ vs. $H_1 : \theta > \theta_0$:

$$T = \frac{\hat{\theta} - \theta_0}{s(\hat{\theta})} \approx Z + \delta, \quad Z \sim N(0, 1)$$

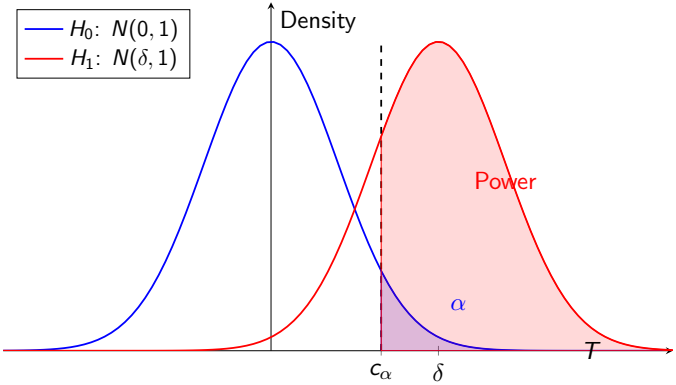
where $\delta = (\theta - \theta_0)/s(\hat{\theta})$ is the **signal-to-noise ratio**.

Power of a one-sided test at level α: $\quad \pi(\delta) = \Phi(\delta - z_\alpha)$

Power increases when:

- The true effect $|\theta - \theta_0|$ is larger
- The standard error $s(\hat{\theta})$ is smaller (more precise estimation)
- The significance level α is larger (less stringent threshold)
- The sample size n is larger (since $s(\hat{\theta}) \propto 1/\sqrt{n}$)

Visualizing Power



Power in OLS: What You Can Control

In OLS, $s(\hat{\theta}) \approx \sigma_e/(\text{sd}(X) \cdot \sqrt{n})$ for a single regressor. So:

$$\delta \approx \frac{(\theta - \theta_0) \cdot \text{sd}(X) \cdot \sqrt{n}}{\sigma_e}$$

Levers for increasing power:

1. **Increase** n: power grows with $\sqrt{n}$
2. **Reduce** σ_e: add controls that explain Y (reduce residual variance)
3. **Increase sd**(X): more variation in the regressor of interest
4. **Increase** α: use 5% instead of 1% (but more Type I errors)

Controls help power! Relevant covariates reduce σ_e without reducing sd(X)—a free power boost. But irrelevant covariates cost degrees of freedom.

Power and Test Dimension

For joint tests ($q > 1$), the Wald statistic under local alternatives: $W \xrightarrow{d} \chi^2_q(\lambda)$, where $\lambda = \boldsymbol{h}' \boldsymbol{V}_\theta^{-1} \boldsymbol{h}$ is the **non-centrality parameter**.

Critical fact: For a fixed λ, power *decreases* as q increases.

q (restrictions)	λ for 50% power	Sample size increase
1	3.85	baseline
2	4.96	+28%
3	5.77	+50%

Takeaway: Testing more restrictions simultaneously dilutes power. A single-coefficient t-test is more powerful than a joint F-test that includes other restrictions.

The 50% Power Benchmark

How far must the true parameter be from θ_0 for 50% power?

One-sided test:

- At $\alpha = 0.05$: need $\delta \geq 1.65$ standard errors
- At $\alpha = 0.01$: need $\delta \geq 2.33$ standard errors

Implication for sample size: $(2.33/1.65)^2 \approx 2$.

> A test at $\alpha = 0.01$ requires roughly **twice the sample size** as $\alpha = 0.05$ to achieve the same power.

Two-sided ($\alpha = 0.05$): need $|\delta| \geq 1.96$ for 50% power. Since se $\propto 1/\sqrt{n}$, you need $n \propto (\sigma_e/(\theta - \theta_0))^2$.

Power Calculation in R

```
# Approximate power for a two-sided t-test in OLS
# Inputs: effect size, residual SD, regressor SD, n, alpha
power_ols <- function(effect, sigma_e, sd_x, n, alpha=0.05) {
  se <- sigma_e / (sd_x * sqrt(n))
  delta <- effect / se
  z <- qnorm(1 - alpha/2)
  power <- pnorm(delta - z) + pnorm(-delta - z)
  return(power)
}

# Example: detect beta = 0.1 with sigma_e = 1, sd(x) = 2
sapply(c(50, 100, 200, 500, 1000), function(n)
  round(power_ols(0.1, 1, 2, n), 3))
# [1] 0.080 0.117 0.198 0.463 0.803
```

With $\beta = 0.1$, $\sigma_e = 1$, $\text{sd}(X) = 2$: you need $n \approx 1000$ for 80% power!

Test Consistency

Definition

A test is **consistent against fixed alternatives** *if for any true* $\theta \neq \theta_0$*:* $P[\text{Reject } H_0 \mid \theta] \to 1$ *as* $n \to \infty$*.*

Good news: The t-test and Wald test are consistent. As $n \to \infty$, $s(\hat{\theta}) \to 0$, so $|T| \to \infty$ whenever $\theta \neq \theta_0$.

Caution: Consistency means you will eventually reject any false null—including *economically trivial* deviations from θ_0.

In very large samples, statistical significance $\neq$ economic significance. A statistically significant but tiny coefficient may not be meaningful.

Hansen's Rules for Applied Work

1. **Report standard errors, not t-ratios.**
   - Standard errors focus attention on precision and confidence intervals.
2. **Report p-values, not asterisks.**
   - p-values contain more information than $*/**/***$ ("an inferior practice").
3. **Focus on economically motivated hypotheses.**
   - Don't mechanically test every coefficient against zero.
   - Report the t-test for $\beta_j = 0$ when this is a scientifically interesting question.

More Practical Advice

4. "Do Not Reject" $\neq$ "Accept."

- Failing to reject means insufficient evidence, *not* that H_0 is true.
- Never write: "the regression finds that X has no effect on Y."

5. Statistical significance $\neq$ economic significance.

- With large n, even tiny effects become "significant."
- Always discuss the *magnitude* and substantive meaning.

6. For nonlinear hypotheses, use minimum distance or test inversion.

- The Wald statistic is *not invariant* to the algebraic formulation of H_0.
- Different equivalent formulations can give different results!

The "What to Report" Checklist

For any coefficient of interest θ:

1. **Point estimate** $\hat{\theta}$: the "best guess"
2. **Standard error** $s(\hat{\theta})$: measure of precision
3. **Confidence interval**: range of values consistent with the data
4. **Economic interpretation**: what the magnitude means in context

When testing:

5. State the **hypothesis** clearly (what question does this answer?)
6. Report the p-**value** (not just "significant" / "not significant")
7. Discuss **power** if you fail to reject (could you have detected a meaningful effect?)
8. Account for **multiple testing** if examining many hypotheses

Summary

Topic	Key Takeaway
F test	Measures fit loss from imposing H_0; $F = W^0/q$; needs homoskedasticity
Score test	Tests from restricted model; equivalent to F for linear restrictions
Trinity	Wald, Score, F agree for linear H_0 in normal model
Test inversion	Confidence sets = non-rejected values; essential for nonlinear parameters
Bonferroni	With k tests, use α/k to control familywise error
Power	$\delta =$ effect/se; grows with $\sqrt{n}$; 50% power needs $\delta \approx 2$
Practical advice	Report SEs, CIs, p-values; focus on magnitudes; consider power

Next Time

Lecture 13: Instrumental Variables

- What happens when $E[\boldsymbol{X}\boldsymbol{e}] \neq \mathbf{0}$?
- Omitted variable bias, simultaneity, measurement error
- The IV solution: find $\boldsymbol{Z}$ such that $E[\boldsymbol{Z}\boldsymbol{e}] = \mathbf{0}$ but $E[\boldsymbol{Z}\boldsymbol{X}'] \neq \mathbf{0}$